# Weekly Analysis Update: August 14, 2026

Chatura Kuruppu

New Mexico State University
SeaQuest Experiment (E906)

August 14, 2026

# Overview

- Review of comments from the recent E906 presentation
- Updated cross-section results
- Evaluation of unfolding systematic uncertainties

# Review of Comments on Unfolded Cross-Section Results (Part 1)

1. **Were Messy/Clean corrections applied during unfolding?**
   Messy/Clean corrections are accounted for during the acceptance correction phase, which is handled separately from the unfolding process.

2. **How should the large standard deviations in the ratio above 7 GeV be interpreted?**
   Error bars were calculated by rigorously propagating uncertainties, including fully correlated systematic effects (which were previously underestimated).

3. **We need to define the systematic uncertainty introduced by unfolding. (Proposed method: calculating the systematic uncertainty by varying the number of iterations by one step.)**
   *Why is Clean unfolding preferred over Messy unfolding?*
   Using Messy MC for unfolding risks double-counting corrections, as the acceptance correction already incorporates Messy MC effects.

# Review of Comments on Unfolded Cross-Section Results (Part 2)

4. **Is it possible to use a different number of iterations for Messy versus Clean unfolding?**
   Currently, we determine the optimal number of iterations by minimizing the divergence between the mean statistical error and the $\Delta\chi^2$ convergence curves.

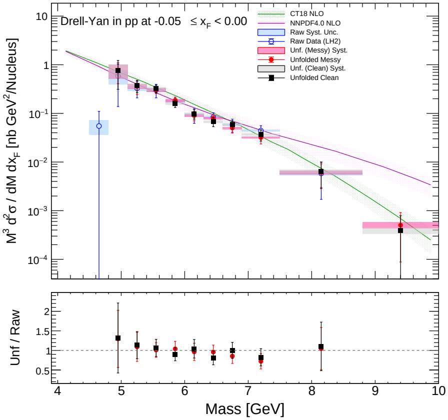
5. **To determine unfolding systematics, could we adopt the closure test methodology used by Harsha?**
   This requires further discussion with the working group.

6. **Combine overflow and underflow bins, and investigate the required bin widths (they should exceed the detector resolution).**
   We plan to broaden the peripheral bins and evaluate the impact on the final cross-section results.

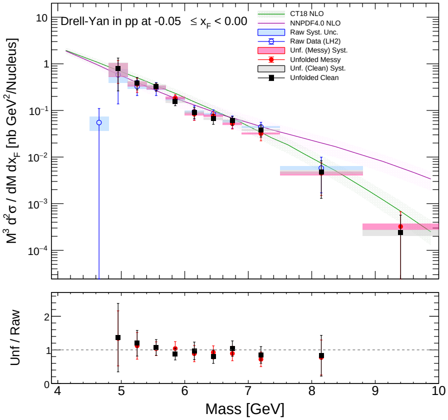
# Updated Cross-Section Results

# LH2 Target: $-0.05 \leq x_F < 0.00$

## Iterations = 3

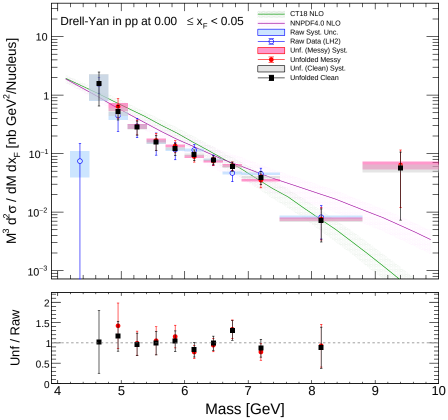



## Iterations = 4

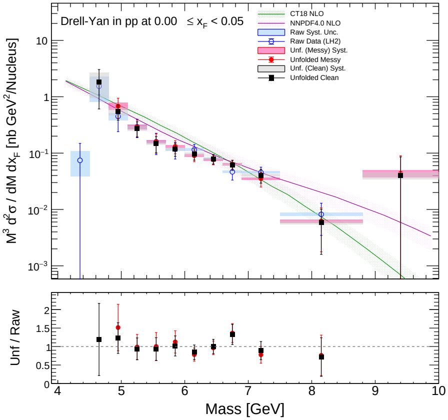


# LH2 Target: $0.00 \leq x_F < 0.05$

## Iterations = 3

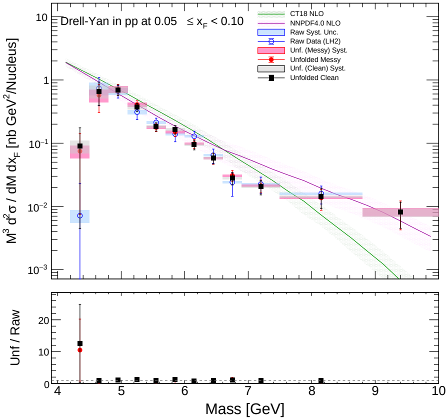



## Iterations = 4

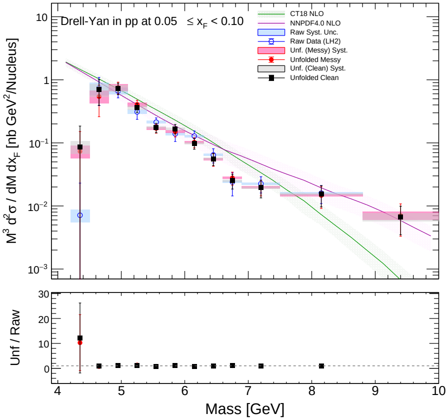


# LH2 Target: $0.05 \leq x_F < 0.10$

## Iterations = 3

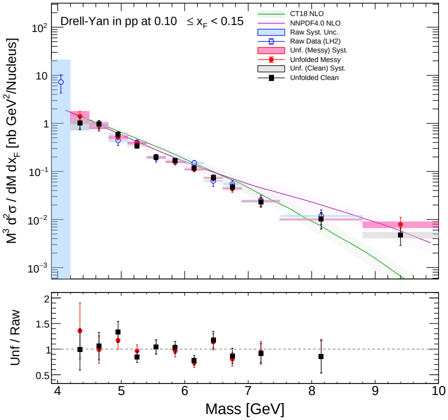



## Iterations = 4

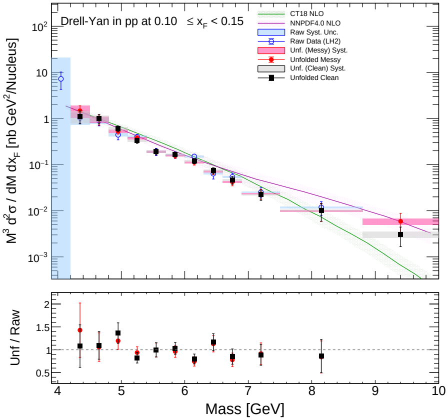


# LH2 Target: $0.10 \leq x_F < 0.15$

## Iterations = 3

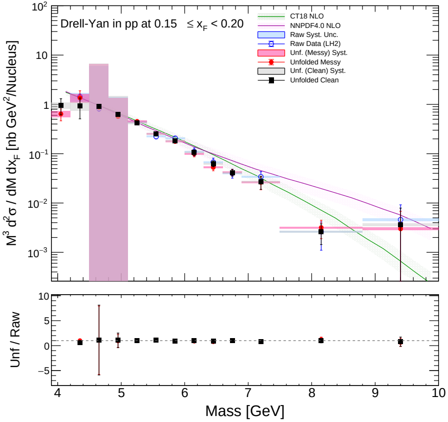



## Iterations = 4

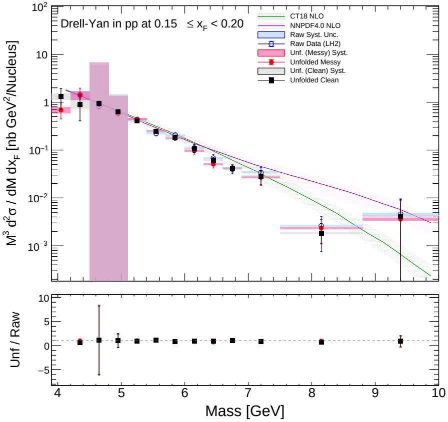


# LH2 Target: $0.15 \le x_F < 0.20$

## Iterations = 3

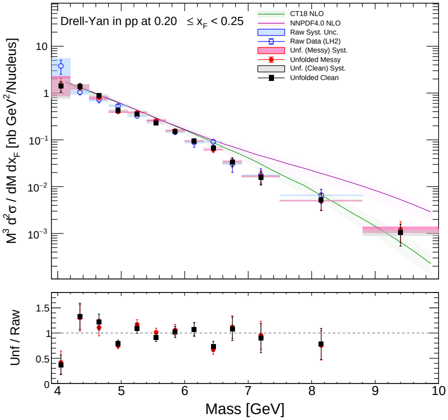



## Iterations = 4

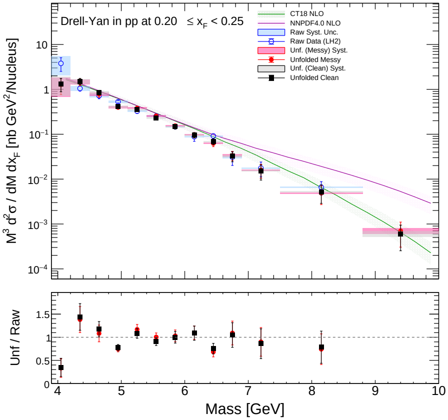


# LH2 Target: $0.20 \leq x_F < 0.25$

**Iterations = 3**

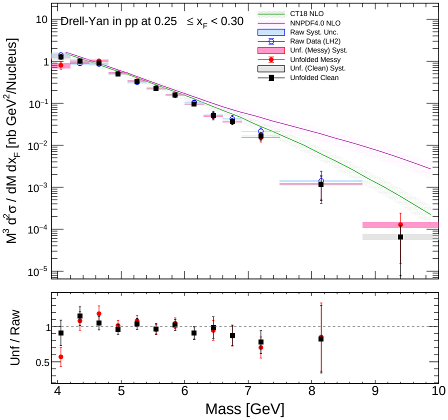



**Iterations = 4**

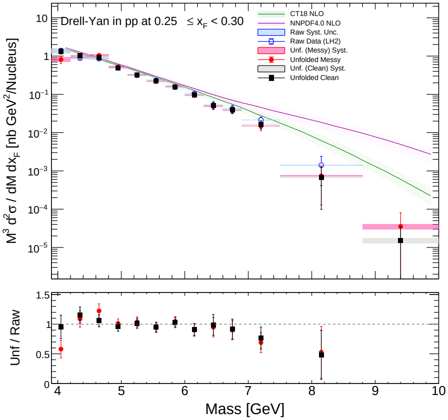


# LH2 Target: $0.25 \leq x_F < 0.30$

## Iterations = 3

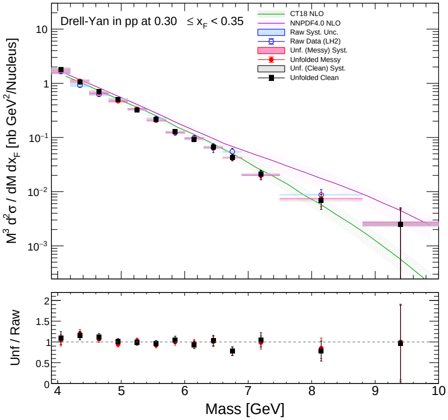



## Iterations = 4

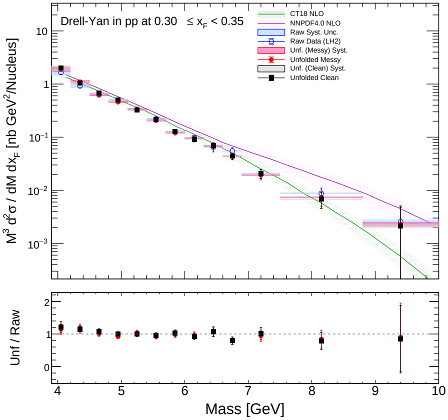


# LH2 Target: $0.30 \leq x_F < 0.35$

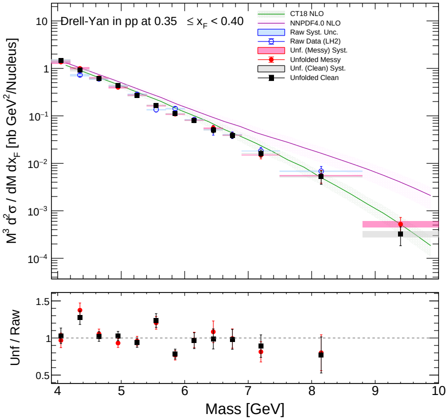
## Iterations = 3



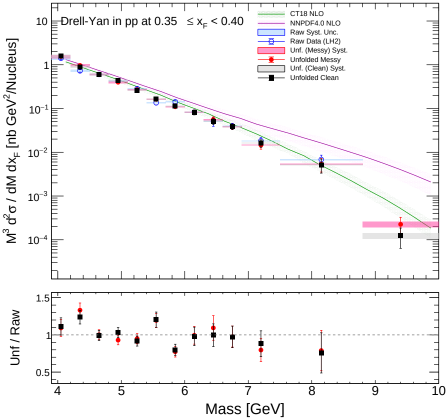

## Iterations = 4



**Iterations = 3**

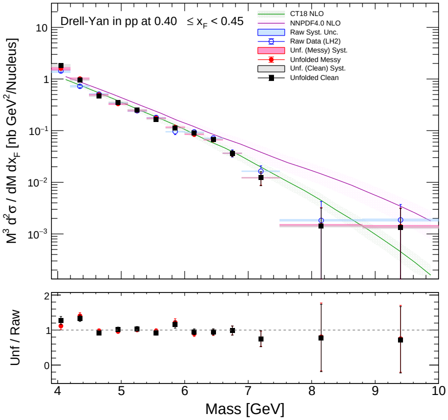



**Iterations = 4**

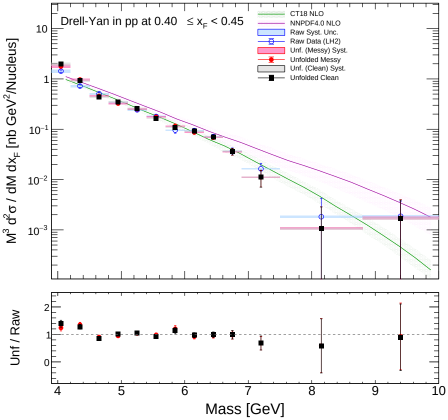


# LH2 Target: $0.40 \leq x_F < 0.45$

## Iterations = 3

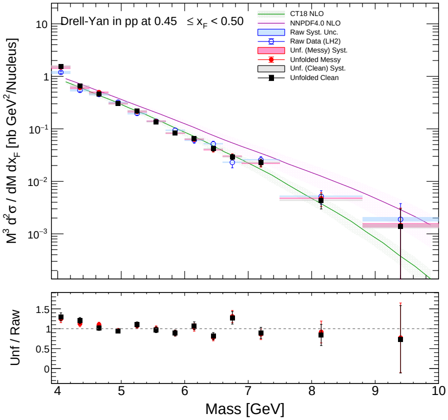



## Iterations = 4

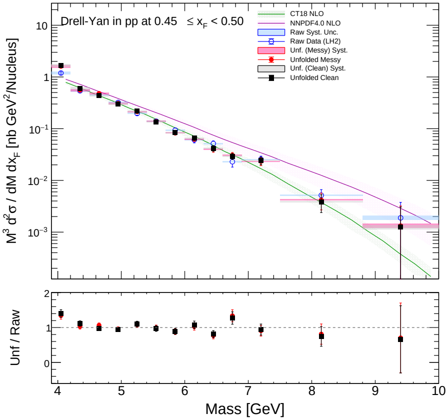


# LH2 Target: $0.45 \leq x_F < 0.50$

## Iterations = 3

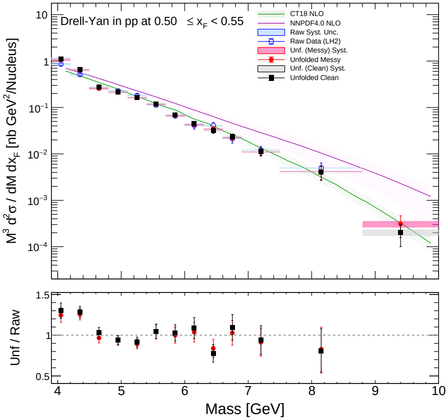



## Iterations = 4

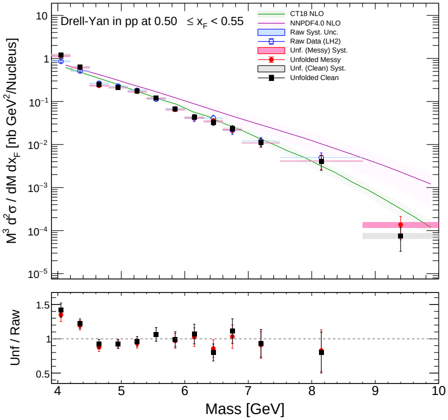


# LH2 Target: $0.50 \leq x_F < 0.55$

## Iterations = 3

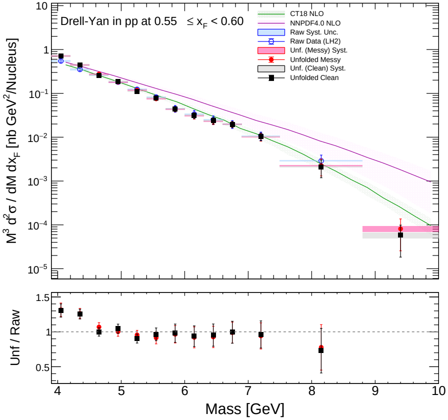



## Iterations = 4

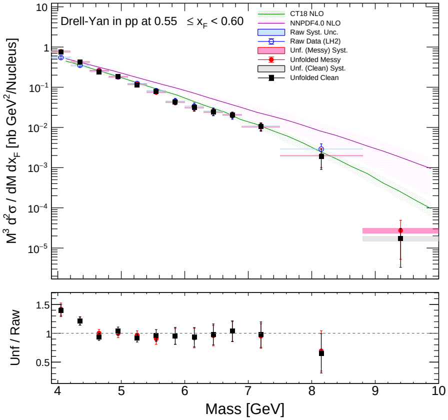


# LH2 Target: $0.55 \leq x_F < 0.60$

## Iterations = 3

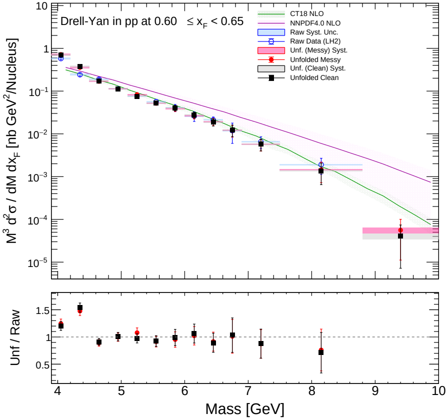



## Iterations = 4

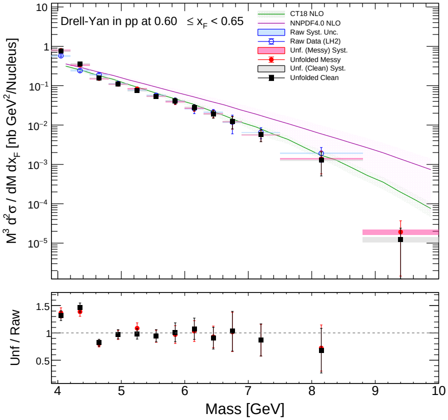


# LH2 Target: $0.60 \leq x_F < 0.65$

**Iterations = 3**

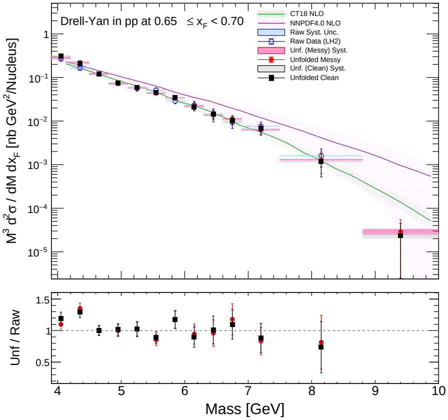



**Iterations = 4**

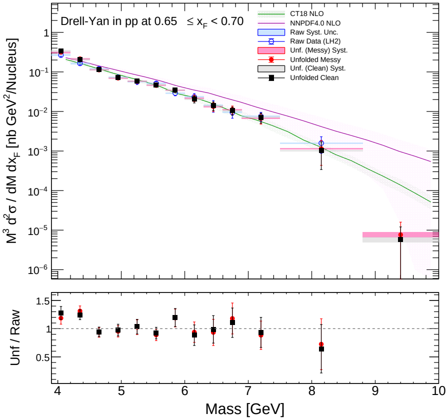


# LH2 Target: $0.65 \leq x_F < 0.70$

## Iterations = 3

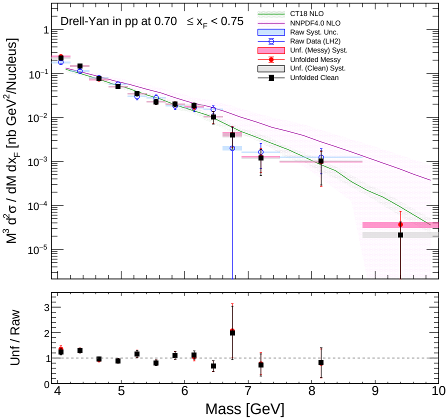



## Iterations = 4

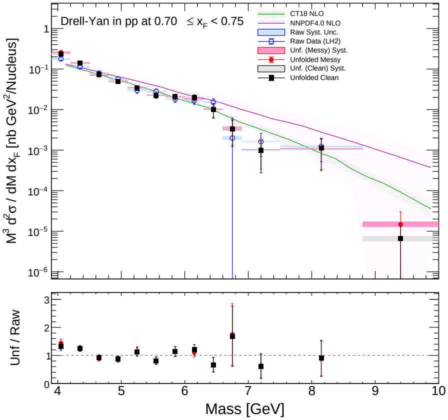


# LH2 Target: $0.70 \leq x_F < 0.75$

## Iterations = 3

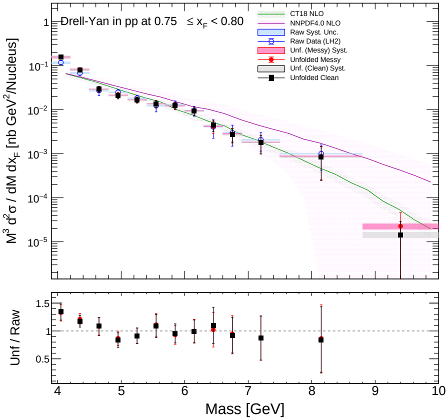



## Iterations = 4

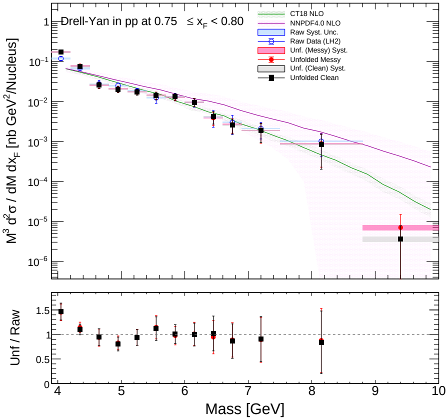


# LH2 Target: $0.75 \leq x_F < 0.80$

## Iterations = 3

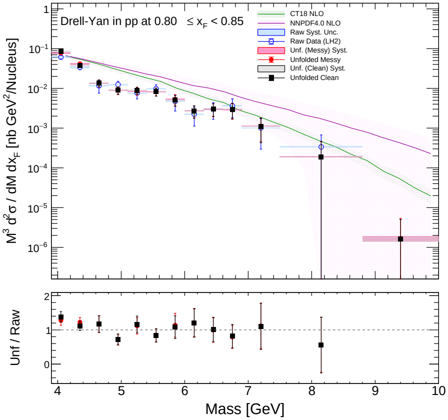



## Iterations = 4

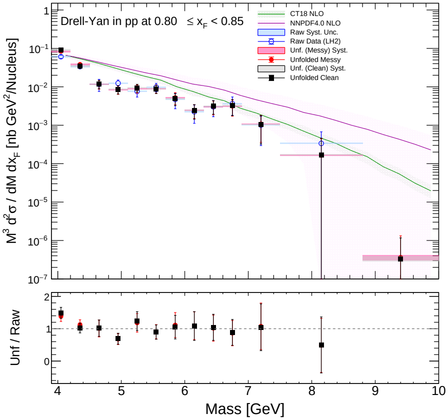


# LH2 Target: $0.80 \leq x_F < 0.85$

## Iterations = 3

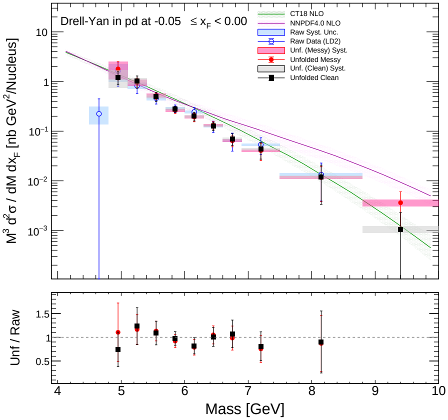



## Iterations = 4

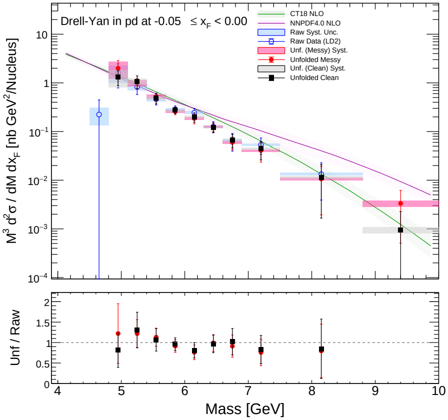


# LD2 Target: $-0.05 \le x_F < 0.00$

## Iterations = 3

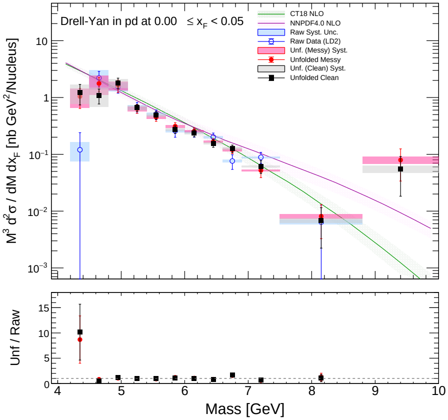



## Iterations = 4

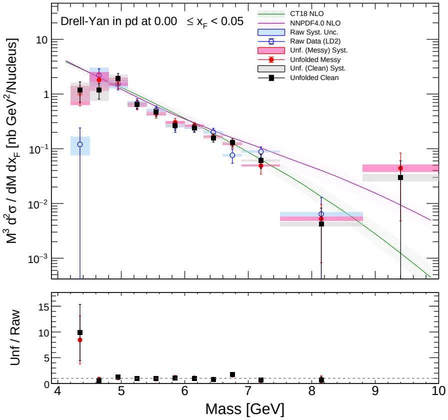


# LD2 Target: $0.00 \leq x_F < 0.05$

**Iterations = 3**

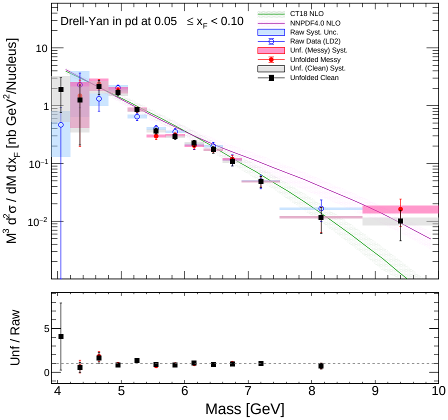



**Iterations = 4**

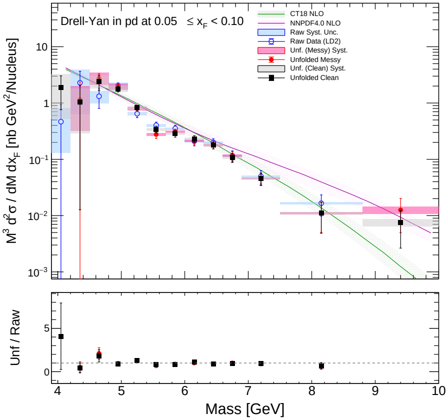


# LD2 Target: $0.05 \leq x_F < 0.10$

## Iterations = 3

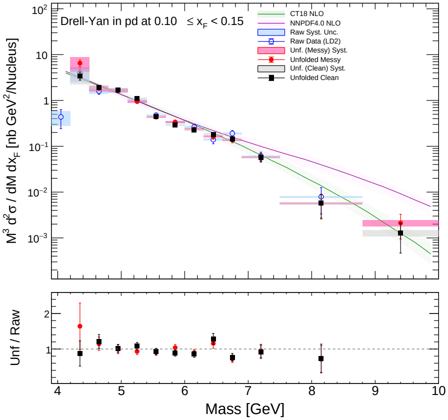



## Iterations = 4

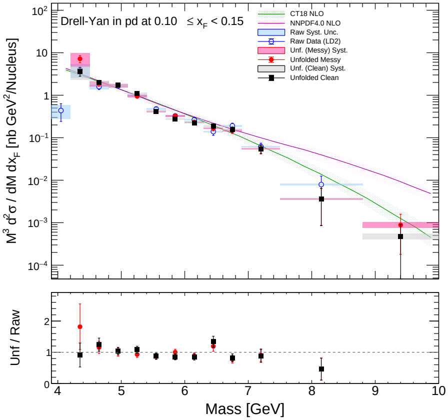


# LD2 Target: $0.10 \leq x_F < 0.15$

## Iterations = 3

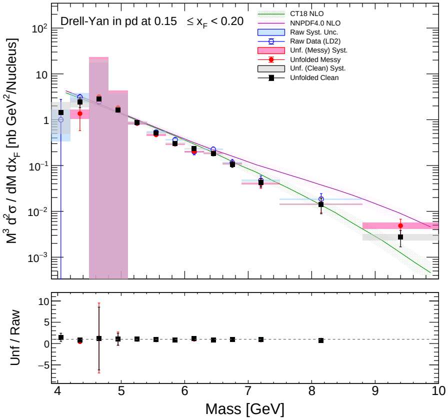



## Iterations = 4

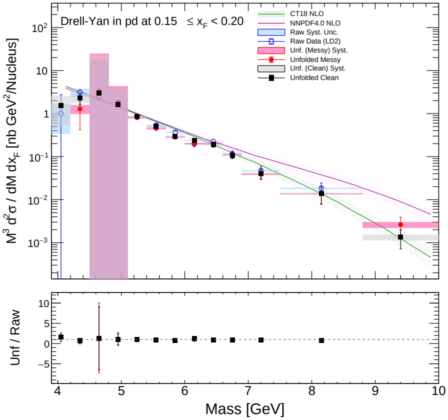


# LD2 Target: $0.15 \leq x_F < 0.20$

## Iterations = 3

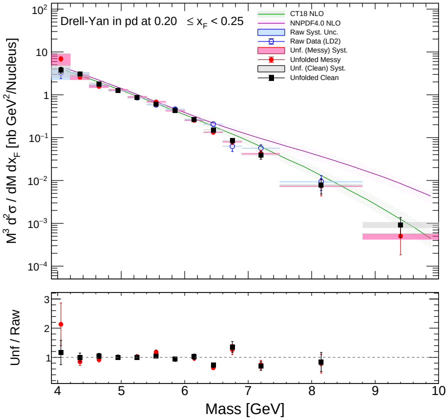



## Iterations = 4

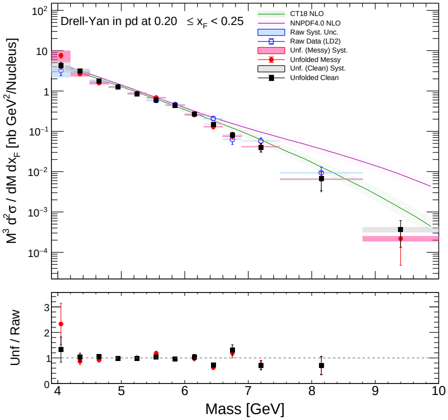


# LD2 Target: $0.20 \leq x_F < 0.25$

**Iterations = 3**

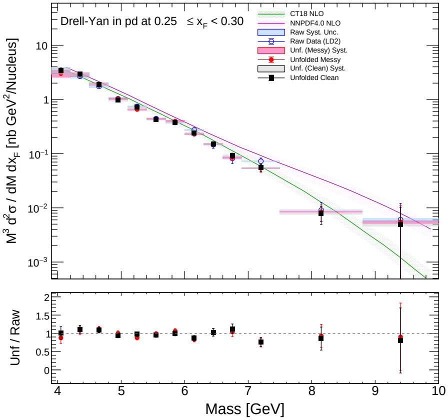



**Iterations = 4**

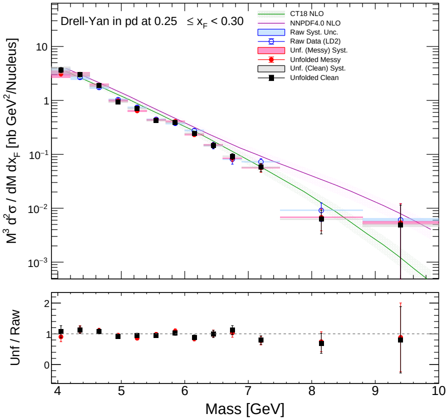


# LD2 Target: $0.25 \leq x_F < 0.30$

## Iterations = 3

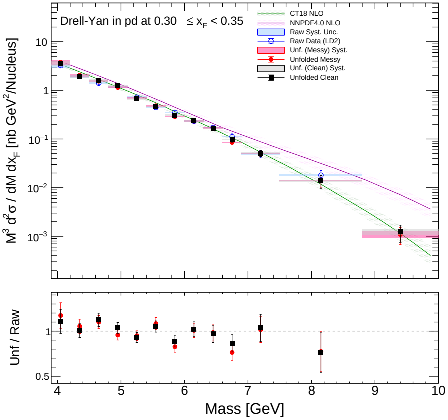



## Iterations = 4

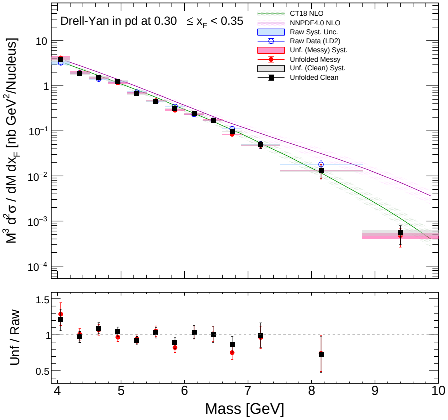


# LD2 Target: $0.30 \leq x_F < 0.35$

## Iterations = 3

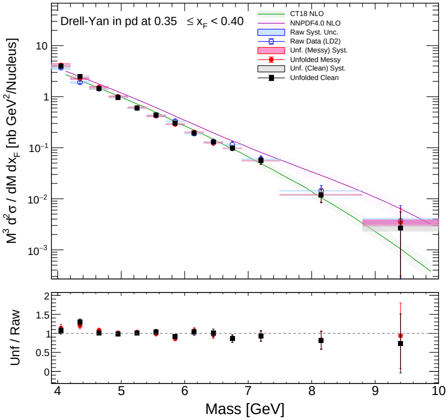



## Iterations = 4

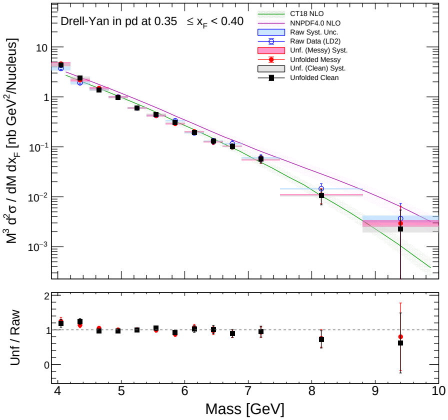


# LD2 Target: $0.35 \leq x_F < 0.40$

## Iterations = 3

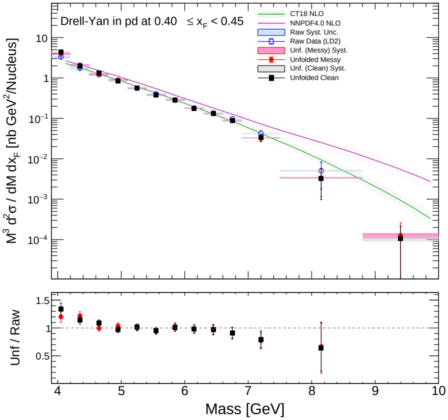



## Iterations = 4

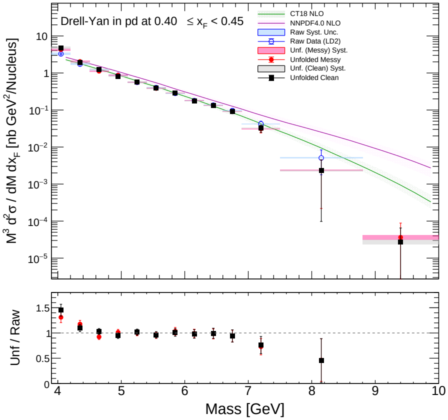


# LD2 Target: $0.40 \leq x_F < 0.45$

## Iterations = 3

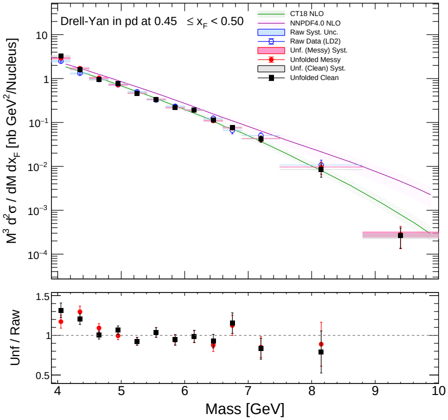



## Iterations = 4

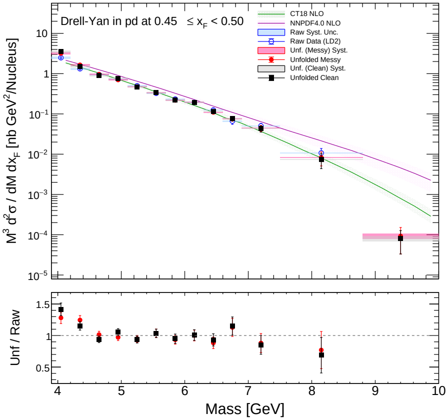


# LD2 Target: $0.45 \le x_F < 0.50$

## Iterations = 3

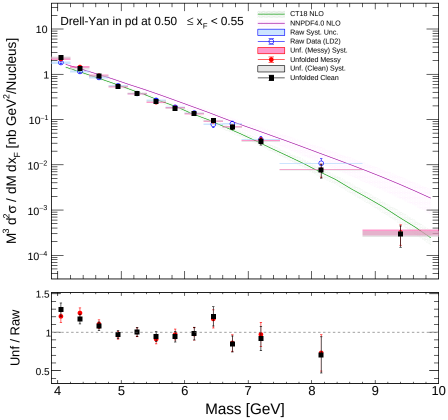



## Iterations = 4

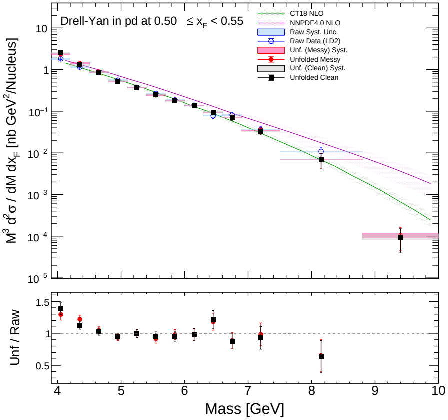


# LD2 Target: $0.50 \leq x_F < 0.55$

## Iterations = 3

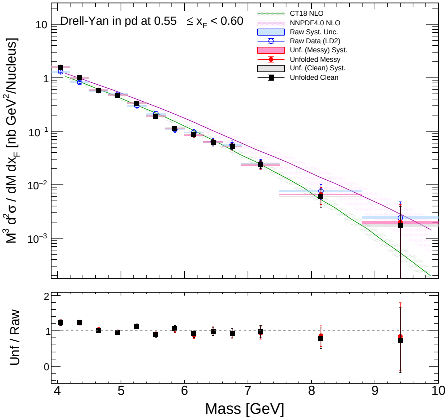



## Iterations = 4

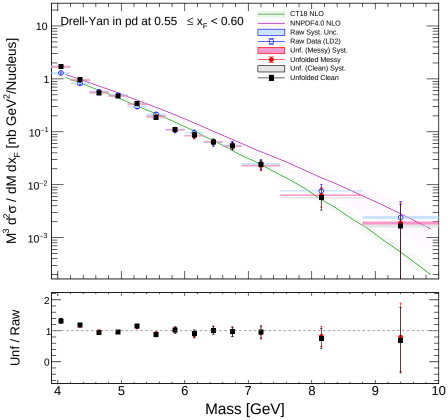


# LD2 Target: $0.55 \leq x_F < 0.60$

## Iterations = 3

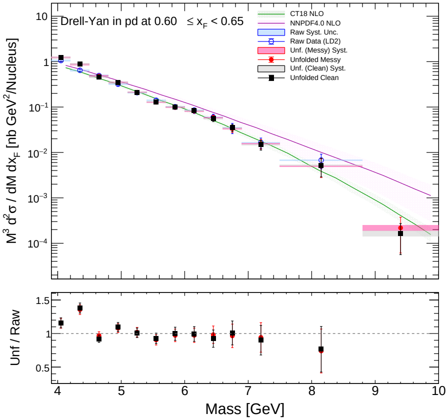



## Iterations = 4

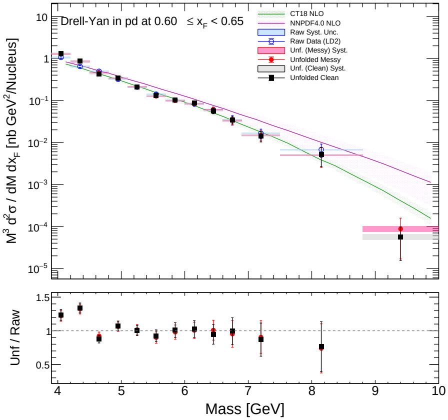


# LD2 Target: $0.60 \le x_F < 0.65$

## Iterations = 3

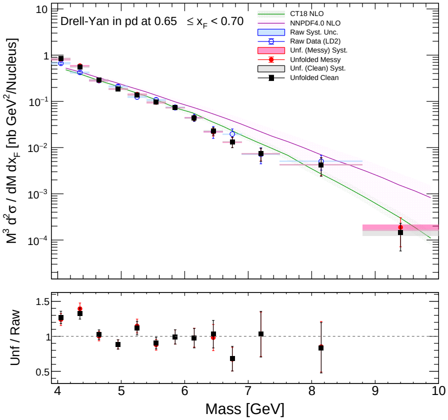



## Iterations = 4

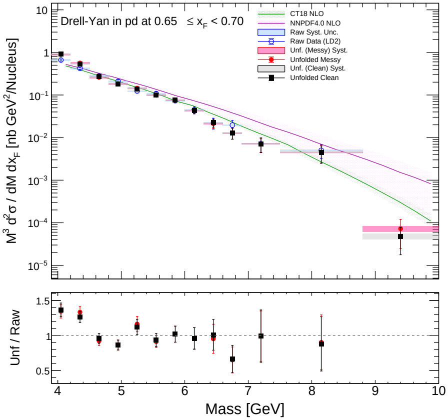


# LD2 Target: $0.65 \leq x_F < 0.70$

## Iterations = 3

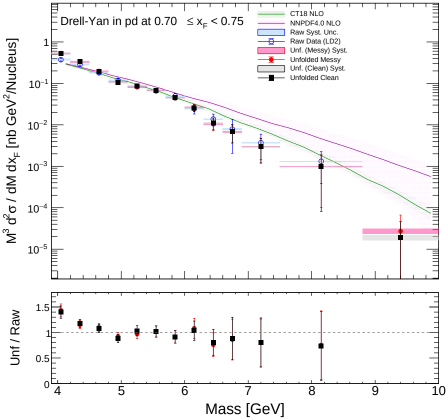



## Iterations = 4

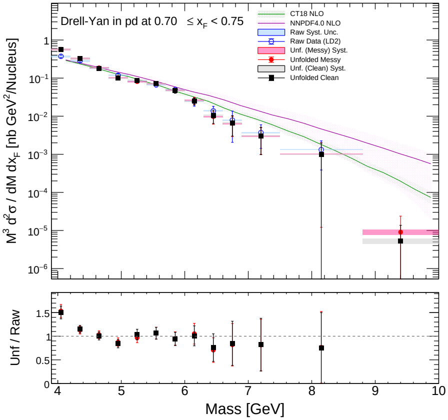


# LD2 Target: $0.70 \leq x_F < 0.75$

## Iterations = 3

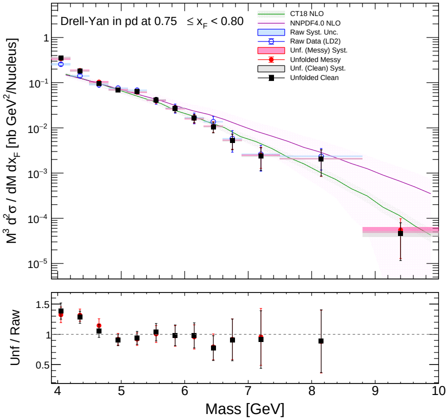



## Iterations = 4

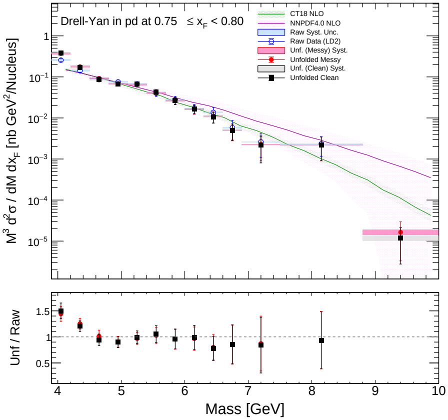


# LD2 Target: $0.75 \leq x_F < 0.80$

**Iterations = 3**

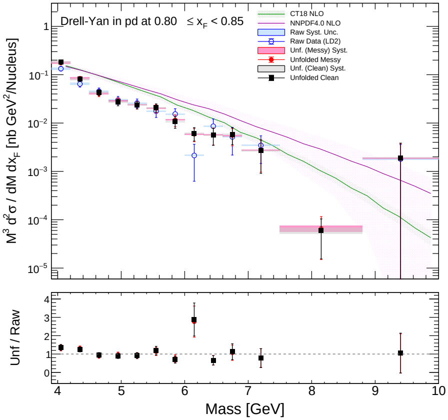



**Iterations = 4**

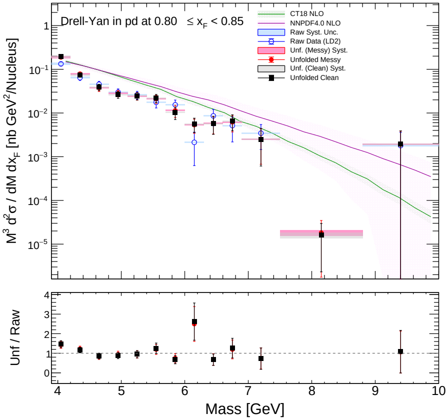


# LD2 Target: $0.80 \leq x_F < 0.85$

**Iterations = 3**




**Iterations = 4**



# Evaluation of Systematic Uncertainty Due to Unfolding

We determine the systematic uncertainty introduced by the unfolding algorithm for both the Messy and Clean configurations.

## Systematic Uncertainty Calculation

$$\text{Systematic Uncertainty } (\%) = \left( \frac{|\sigma_{\text{unf}}^{N=4} - \sigma_{\text{unf}}^{N=3}|}{\sigma_{\text{unf}}^{N=3}} \right) \times 100$$

*Note: The deviation between consecutive algorithmic iterations provides a robust estimate of the structural bias introduced by the unfolding matrix.*

# Unfolding Systematics: $-0.05 \leq x_F < 0.00$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $4.50 \leq$ Mass $< 4.80$ | - | - |
| $4.80 \leq$ Mass $< 5.10$ | 3.55 | 3.75 |
| $5.10 \leq$ Mass $< 5.40$ | 5.51 | 2.77 |
| $5.40 \leq$ Mass $< 5.70$ | 0.66 | 2.57 |
| $5.70 \leq$ Mass $< 6.00$ | 2.82 | 0.05 |
| $6.00 \leq$ Mass $< 6.30$ | 6.88 | 7.88 |
| $6.30 \leq$ Mass $< 6.60$ | 1.48 | 3.55 |
| $6.60 \leq$ Mass $< 6.90$ | 3.75 | 4.40 |
| $6.90 \leq$ Mass $< 7.50$ | 3.06 | 0.38 |
| $7.50 \leq$ Mass $< 8.80$ | 25.12 | 25.98 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $4.50 \leq$ Mass $< 4.80$ | - | - |
| $4.80 \leq$ Mass $< 5.10$ | 9.31 | 10.70 |
| $5.10 \leq$ Mass $< 5.40$ | 6.00 | 4.97 |
| $5.40 \leq$ Mass $< 5.70$ | 2.06 | 0.03 |
| $5.70 \leq$ Mass $< 6.00$ | 1.48 | 0.48 |
| $6.00 \leq$ Mass $< 6.30$ | 0.81 | 3.34 |
| $6.30 \leq$ Mass $< 6.60$ | 3.88 | 5.63 |
| $6.60 \leq$ Mass $< 6.90$ | 4.62 | 6.99 |
| $6.90 \leq$ Mass $< 7.50$ | 2.37 | 0.43 |
| $7.50 \leq$ Mass $< 8.80$ | 5.79 | 8.38 |

# Unfolding Systematics: $0.00 \leq x_F < 0.05$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $4.20 \leq$ Mass $< 4.50$ | - | - |
| $4.50 \leq$ Mass $< 4.80$ | 16.49 | - |
| $4.80 \leq$ Mass $< 5.10$ | 5.66 | 6.63 |
| $5.10 \leq$ Mass $< 5.40$ | 2.81 | 0.37 |
| $5.40 \leq$ Mass $< 5.70$ | 6.20 | 4.78 |
| $5.70 \leq$ Mass $< 6.00$ | 2.41 | 2.58 |
| $6.00 \leq$ Mass $< 6.30$ | 1.76 | 0.99 |
| $6.30 \leq$ Mass $< 6.60$ | 0.96 | 2.02 |
| $6.60 \leq$ Mass $< 6.90$ | 1.80 | 2.41 |
| $6.90 \leq$ Mass $< 7.50$ | 2.45 | 0.43 |
| $7.50 \leq$ Mass $< 8.80$ | 18.71 | 18.03 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $4.20 \leq$ Mass $< 4.50$ | 2.73 | 2.68 |
| $4.50 \leq$ Mass $< 4.80$ | 9.75 | 2.46 |
| $4.80 \leq$ Mass $< 5.10$ | 6.64 | 4.62 |
| $5.10 \leq$ Mass $< 5.40$ | 1.91 | 0.92 |
| $5.40 \leq$ Mass $< 5.70$ | 3.41 | 2.22 |
| $5.70 \leq$ Mass $< 6.00$ | 3.67 | 2.47 |
| $6.00 \leq$ Mass $< 6.30$ | 1.97 | 2.49 |
| $6.30 \leq$ Mass $< 6.60$ | 1.07 | 1.04 |
| $6.60 \leq$ Mass $< 6.90$ | 3.16 | 2.40 |
| $6.90 \leq$ Mass $< 7.50$ | 1.44 | 5.79 |
| $7.50 \leq$ Mass $< 8.80$ | 38.96 | 35.50 |

# Unfolding Systematics: $0.05 \leq x_F < 0.10$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $4.20 \leq$ Mass $< 4.50$ | 3.28 | 2.01 |
| $4.50 \leq$ Mass $< 4.80$ | 6.03 | 5.18 |
| $4.80 \leq$ Mass $< 5.10$ | 5.05 | 6.55 |
| $5.10 \leq$ Mass $< 5.40$ | 2.46 | 0.31 |
| $5.40 \leq$ Mass $< 5.70$ | 4.48 | 4.11 |
| $5.70 \leq$ Mass $< 6.00$ | 1.58 | 0.19 |
| $6.00 \leq$ Mass $< 6.30$ | 3.25 | 3.94 |
| $6.30 \leq$ Mass $< 6.60$ | 4.19 | 4.86 |
| $6.60 \leq$ Mass $< 6.90$ | 10.33 | 9.69 |
| $6.90 \leq$ Mass $< 7.50$ | 4.77 | 6.04 |
| $7.50 \leq$ Mass $< 8.80$ | 7.34 | 7.81 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 0.00 | - |
| $4.20 \leq$ Mass $< 4.50$ | 16.81 | 22.73 |
| $4.50 \leq$ Mass $< 4.80$ | 11.53 | 15.91 |
| $4.80 \leq$ Mass $< 5.10$ | 5.84 | 5.68 |
| $5.10 \leq$ Mass $< 5.40$ | 2.89 | 6.38 |
| $5.40 \leq$ Mass $< 5.70$ | 7.01 | 7.02 |
| $5.70 \leq$ Mass $< 6.00$ | 2.67 | 0.40 |
| $6.00 \leq$ Mass $< 6.30$ | 2.98 | 4.39 |
| $6.30 \leq$ Mass $< 6.60$ | 3.87 | 2.32 |
| $6.60 \leq$ Mass $< 6.90$ | 0.53 | 2.39 |
| $6.90 \leq$ Mass $< 7.50$ | 7.07 | 7.49 |
| $7.50 \leq$ Mass $< 8.80$ | 5.06 | 5.58 |

# Unfolding Systematics: $0.10 \leq x_F < 0.15$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | - | - |
| $4.20 \leq$ Mass $< 4.50$ | 8.54 | 5.39 |
| $4.50 \leq$ Mass $< 4.80$ | 3.55 | 6.83 |
| $4.80 \leq$ Mass $< 5.10$ | 2.20 | 2.09 |
| $5.10 \leq$ Mass $< 5.40$ | 2.78 | 2.70 |
| $5.40 \leq$ Mass $< 5.70$ | 4.26 | 4.41 |
| $5.70 \leq$ Mass $< 6.00$ | 0.11 | 0.33 |
| $6.00 \leq$ Mass $< 6.30$ | 2.21 | 1.42 |
| $6.30 \leq$ Mass $< 6.60$ | 1.33 | 0.85 |
| $6.60 \leq$ Mass $< 6.90$ | 2.17 | 2.89 |
| $6.90 \leq$ Mass $< 7.50$ | 2.40 | 2.90 |
| $7.50 \leq$ Mass $< 8.80$ | 0.29 | 0.71 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | - | - |
| $4.20 \leq$ Mass $< 4.50$ | 4.42 | 10.66 |
| $4.50 \leq$ Mass $< 4.80$ | 3.01 | 0.56 |
| $4.80 \leq$ Mass $< 5.10$ | 2.20 | 1.55 |
| $5.10 \leq$ Mass $< 5.40$ | 0.06 | 0.03 |
| $5.40 \leq$ Mass $< 5.70$ | 5.87 | 5.31 |
| $5.70 \leq$ Mass $< 6.00$ | 4.06 | 3.70 |
| $6.00 \leq$ Mass $< 6.30$ | 0.66 | 1.72 |
| $6.30 \leq$ Mass $< 6.60$ | 4.06 | 2.87 |
| $6.60 \leq$ Mass $< 6.90$ | 7.21 | 7.43 |
| $6.90 \leq$ Mass $< 7.50$ | 4.43 | 4.51 |
| $7.50 \leq$ Mass $< 8.80$ | 37.79 | 35.12 |

# Unfolding Systematics: $0.15 \leq x_F < 0.20$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $4.20 \leq$ Mass $< 4.50$ | 1.78 | 1.94 |
| $4.50 \leq$ Mass $< 4.80$ | 3.55 | 2.95 |
| $4.80 \leq$ Mass $< 5.10$ | 0.84 | 0.93 |
| $5.10 \leq$ Mass $< 5.40$ | 3.04 | 1.53 |
| $5.40 \leq$ Mass $< 5.70$ | 2.65 | 2.00 |
| $5.70 \leq$ Mass $< 6.00$ | 0.93 | 0.20 |
| $6.00 \leq$ Mass $< 6.30$ | 0.31 | 2.77 |
| $6.30 \leq$ Mass $< 6.60$ | 3.56 | 3.91 |
| $6.60 \leq$ Mass $< 6.90$ | 0.67 | 1.77 |
| $6.90 \leq$ Mass $< 7.50$ | 4.11 | 3.59 |
| $7.50 \leq$ Mass $< 8.80$ | 29.95 | 26.68 |
| $8.80 \leq$ Mass $< 10.00$ | 14.81 | 20.32 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.52 | - |
| $4.20 \leq$ Mass $< 4.50$ | 4.09 | 5.22 |
| $4.50 \leq$ Mass $< 4.80$ | 5.84 | 5.34 |
| $4.80 \leq$ Mass $< 5.10$ | 0.73 | 0.43 |
| $5.10 \leq$ Mass $< 5.40$ | 3.36 | 3.14 |
| $5.40 \leq$ Mass $< 5.70$ | 2.07 | 3.09 |
| $5.70 \leq$ Mass $< 6.00$ | 2.07 | 2.80 |
| $6.00 \leq$ Mass $< 6.30$ | 1.21 | 0.54 |
| $6.30 \leq$ Mass $< 6.60$ | 3.76 | 5.28 |
| $6.60 \leq$ Mass $< 6.90$ | 0.86 | 1.79 |
| $6.90 \leq$ Mass $< 7.50$ | 4.92 | 3.67 |
| $7.50 \leq$ Mass $< 8.80$ | 2.78 | 4.40 |

# Unfolding Systematics: $0.20 \leq x_F < 0.25$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq \text{Mass} < 4.20$ | 5.87 | 20.21 |
| $4.20 \leq \text{Mass} < 4.50$ | 8.16 | 6.27 |
| $4.50 \leq \text{Mass} < 4.80$ | 3.58 | 2.42 |
| $4.80 \leq \text{Mass} < 5.10$ | 2.17 | 0.33 |
| $5.10 \leq \text{Mass} < 5.40$ | 0.59 | 0.12 |
| $5.40 \leq \text{Mass} < 5.70$ | 0.54 | 0.77 |
| $5.70 \leq \text{Mass} < 6.00$ | 1.97 | 2.54 |
| $6.00 \leq \text{Mass} < 6.30$ | 1.57 | 2.02 |
| $6.30 \leq \text{Mass} < 6.60$ | 2.38 | 1.26 |
| $6.60 \leq \text{Mass} < 6.90$ | 3.37 | 2.33 |
| $6.90 \leq \text{Mass} < 7.50$ | 4.14 | 5.79 |
| $7.50 \leq \text{Mass} < 8.80$ | 0.01 | 1.77 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq \text{Mass} < 4.20$ | 13.87 | 9.26 |
| $4.20 \leq \text{Mass} < 4.50$ | 3.32 | 2.14 |
| $4.50 \leq \text{Mass} < 4.80$ | 0.06 | 0.03 |
| $4.80 \leq \text{Mass} < 5.10$ | 1.78 | 2.11 |
| $5.10 \leq \text{Mass} < 5.40$ | 2.46 | 1.97 |
| $5.40 \leq \text{Mass} < 5.70$ | 1.05 | 0.18 |
| $5.70 \leq \text{Mass} < 6.00$ | 1.34 | 2.31 |
| $6.00 \leq \text{Mass} < 6.30$ | 1.69 | 0.54 |
| $6.30 \leq \text{Mass} < 6.60$ | 1.54 | 3.38 |
| $6.60 \leq \text{Mass} < 6.90$ | 3.57 | 5.20 |
| $6.90 \leq \text{Mass} < 7.50$ | 0.99 | 1.26 |
| $7.50 \leq \text{Mass} < 8.80$ | 15.43 | 12.24 |

# Unfolding Systematics: $0.25 \leq x_F < 0.30$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 4.49 | 1.59 |
| $4.20 \leq$ Mass $< 4.50$ | 0.46 | 0.56 |
| $4.50 \leq$ Mass $< 4.80$ | 0.83 | 3.21 |
| $4.80 \leq$ Mass $< 5.10$ | 0.57 | 0.64 |
| $5.10 \leq$ Mass $< 5.40$ | 2.64 | 4.07 |
| $5.40 \leq$ Mass $< 5.70$ | 1.72 | 2.27 |
| $5.70 \leq$ Mass $< 6.00$ | 0.17 | 0.10 |
| $6.00 \leq$ Mass $< 6.30$ | 0.25 | 0.67 |
| $6.30 \leq$ Mass $< 6.60$ | 0.17 | 0.13 |
| $6.60 \leq$ Mass $< 6.90$ | 4.58 | 3.91 |
| $6.90 \leq$ Mass $< 7.50$ | 1.53 | 2.03 |
| $7.50 \leq$ Mass $< 8.80$ | 41.29 | 38.30 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 6.40 | 2.92 |
| $4.20 \leq$ Mass $< 4.50$ | 2.37 | 5.60 |
| $4.50 \leq$ Mass $< 4.80$ | 1.22 | 1.13 |
| $4.80 \leq$ Mass $< 5.10$ | 3.52 | 4.97 |
| $5.10 \leq$ Mass $< 5.40$ | 2.28 | 2.34 |
| $5.40 \leq$ Mass $< 5.70$ | 0.69 | 0.38 |
| $5.70 \leq$ Mass $< 6.00$ | 2.68 | 2.47 |
| $6.00 \leq$ Mass $< 6.30$ | 1.71 | 0.82 |
| $6.30 \leq$ Mass $< 6.60$ | 3.02 | 1.90 |
| $6.60 \leq$ Mass $< 6.90$ | 0.43 | 0.78 |
| $6.90 \leq$ Mass $< 7.50$ | 3.82 | 3.96 |
| $7.50 \leq$ Mass $< 8.80$ | 19.53 | 20.18 |
| $8.80 \leq$ Mass $< 10.00$ | 0.04 | 1.99 |

# Unfolding Systematics: $0.30 \leq x_F < 0.35$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 10.70 | 10.03 |
| $4.20 \leq$ Mass $< 4.50$ | 1.99 | 0.26 |
| $4.50 \leq$ Mass $< 4.80$ | 4.08 | 5.68 |
| $4.80 \leq$ Mass $< 5.10$ | 1.08 | 2.48 |
| $5.10 \leq$ Mass $< 5.40$ | 0.79 | 1.12 |
| $5.40 \leq$ Mass $< 5.70$ | 0.52 | 0.41 |
| $5.70 \leq$ Mass $< 6.00$ | 2.05 | 2.25 |
| $6.00 \leq$ Mass $< 6.30$ | 0.62 | 0.38 |
| $6.30 \leq$ Mass $< 6.60$ | 4.00 | 3.98 |
| $6.60 \leq$ Mass $< 6.90$ | 2.01 | 2.60 |
| $6.90 \leq$ Mass $< 7.50$ | 3.94 | 4.37 |
| $7.50 \leq$ Mass $< 8.80$ | 0.29 | 0.86 |
| $8.80 \leq$ Mass $< 10.00$ | 12.68 | 9.95 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 9.10 | 9.84 |
| $4.20 \leq$ Mass $< 4.50$ | 3.39 | 4.39 |
| $4.50 \leq$ Mass $< 4.80$ | 2.91 | 2.61 |
| $4.80 \leq$ Mass $< 5.10$ | 0.95 | 1.15 |
| $5.10 \leq$ Mass $< 5.40$ | 0.95 | 0.76 |
| $5.40 \leq$ Mass $< 5.70$ | 2.58 | 2.94 |
| $5.70 \leq$ Mass $< 6.00$ | 0.16 | 0.30 |
| $6.00 \leq$ Mass $< 6.30$ | 1.88 | 2.55 |
| $6.30 \leq$ Mass $< 6.60$ | 3.39 | 3.09 |
| $6.60 \leq$ Mass $< 6.90$ | 0.48 | 1.26 |
| $6.90 \leq$ Mass $< 7.50$ | 3.60 | 5.02 |
| $7.50 \leq$ Mass $< 8.80$ | 5.13 | 3.97 |

# Unfolding Systematics: $0.35 \leq x_F < 0.40$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 8.16 | 12.90 |
| $4.20 \leq$ Mass $< 4.50$ | 2.81 | 3.10 |
| $4.50 \leq$ Mass $< 4.80$ | 2.72 | 4.98 |
| $4.80 \leq$ Mass $< 5.10$ | 0.56 | 0.20 |
| $5.10 \leq$ Mass $< 5.40$ | 2.11 | 1.01 |
| $5.40 \leq$ Mass $< 5.70$ | 2.44 | 0.89 |
| $5.70 \leq$ Mass $< 6.00$ | 2.07 | 0.84 |
| $6.00 \leq$ Mass $< 6.30$ | 1.72 | 2.08 |
| $6.30 \leq$ Mass $< 6.60$ | 0.82 | 1.00 |
| $6.60 \leq$ Mass $< 6.90$ | 0.85 | 1.50 |
| $6.90 \leq$ Mass $< 7.50$ | 1.05 | 2.12 |
| $7.50 \leq$ Mass $< 8.80$ | 1.74 | 1.47 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 10.00 | 10.35 |
| $4.20 \leq$ Mass $< 4.50$ | 3.87 | 5.10 |
| $4.50 \leq$ Mass $< 4.80$ | 4.68 | 3.20 |
| $4.80 \leq$ Mass $< 5.10$ | 1.09 | 1.37 |
| $5.10 \leq$ Mass $< 5.40$ | 1.20 | 2.41 |
| $5.40 \leq$ Mass $< 5.70$ | 0.56 | 0.40 |
| $5.70 \leq$ Mass $< 6.00$ | 1.42 | 1.29 |
| $6.00 \leq$ Mass $< 6.30$ | 0.70 | 0.10 |
| $6.30 \leq$ Mass $< 6.60$ | 0.18 | 0.94 |
| $6.60 \leq$ Mass $< 6.90$ | 3.91 | 3.31 |
| $6.90 \leq$ Mass $< 7.50$ | 1.59 | 1.10 |
| $7.50 \leq$ Mass $< 8.80$ | 10.52 | 9.12 |
| $8.80 \leq$ Mass $< 10.00$ | 15.55 | 14.46 |

# Unfolding Systematics: $0.40 \leq x_F < 0.45$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 9.02 | 10.93 |
| $4.20 \leq$ Mass $< 4.50$ | 4.39 | 4.05 |
| $4.50 \leq$ Mass $< 4.80$ | 6.52 | 7.53 |
| $4.80 \leq$ Mass $< 5.10$ | 0.74 | 1.79 |
| $5.10 \leq$ Mass $< 5.40$ | 2.02 | 2.35 |
| $5.40 \leq$ Mass $< 5.70$ | 0.16 | 0.81 |
| $5.70 \leq$ Mass $< 6.00$ | 2.07 | 1.11 |
| $6.00 \leq$ Mass $< 6.30$ | 2.35 | 3.08 |
| $6.30 \leq$ Mass $< 6.60$ | 4.19 | 3.61 |
| $6.60 \leq$ Mass $< 6.90$ | 0.23 | 1.38 |
| $6.90 \leq$ Mass $< 7.50$ | 8.66 | 10.60 |
| $7.50 \leq$ Mass $< 8.80$ | 24.85 | 26.33 |
| $8.80 \leq$ Mass $< 10.00$ | 23.89 | 23.36 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 8.21 | 9.37 |
| $4.20 \leq$ Mass $< 4.50$ | 4.04 | 3.63 |
| $4.50 \leq$ Mass $< 4.80$ | 5.30 | 7.27 |
| $4.80 \leq$ Mass $< 5.10$ | 2.41 | 2.42 |
| $5.10 \leq$ Mass $< 5.40$ | 0.08 | 0.53 |
| $5.40 \leq$ Mass $< 5.70$ | 1.41 | 1.28 |
| $5.70 \leq$ Mass $< 6.00$ | 0.49 | 0.55 |
| $6.00 \leq$ Mass $< 6.30$ | 0.09 | 0.64 |
| $6.30 \leq$ Mass $< 6.60$ | 2.12 | 2.23 |
| $6.60 \leq$ Mass $< 6.90$ | 3.63 | 2.91 |
| $6.90 \leq$ Mass $< 7.50$ | 4.24 | 5.41 |
| $7.50 \leq$ Mass $< 8.80$ | 29.36 | 30.42 |

# Unfolding Systematics: $0.45 \leq x_F < 0.50$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 8.01 | 7.89 |
| $4.20 \leq$ Mass $< 4.50$ | 7.29 | 8.44 |
| $4.50 \leq$ Mass $< 4.80$ | 3.95 | 3.16 |
| $4.80 \leq$ Mass $< 5.10$ | 0.00 | 0.68 |
| $5.10 \leq$ Mass $< 5.40$ | 0.30 | 0.48 |
| $5.40 \leq$ Mass $< 5.70$ | 0.23 | 0.75 |
| $5.70 \leq$ Mass $< 6.00$ | 0.01 | 0.87 |
| $6.00 \leq$ Mass $< 6.30$ | 0.31 | 0.19 |
| $6.30 \leq$ Mass $< 6.60$ | 0.33 | 1.15 |
| $6.60 \leq$ Mass $< 6.90$ | 0.02 | 2.48 |
| $6.90 \leq$ Mass $< 7.50$ | 5.18 | 5.39 |
| $7.50 \leq$ Mass $< 8.80$ | 11.65 | 11.48 |
| $8.80 \leq$ Mass $< 10.00$ | 10.11 | 9.01 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.63 | 9.24 |
| $4.20 \leq$ Mass $< 4.50$ | 4.33 | 3.94 |
| $4.50 \leq$ Mass $< 4.80$ | 6.54 | 7.21 |
| $4.80 \leq$ Mass $< 5.10$ | 1.09 | 2.32 |
| $5.10 \leq$ Mass $< 5.40$ | 1.13 | 2.37 |
| $5.40 \leq$ Mass $< 5.70$ | 0.14 | 0.02 |
| $5.70 \leq$ Mass $< 6.00$ | 0.61 | 0.43 |
| $6.00 \leq$ Mass $< 6.30$ | 1.97 | 1.76 |
| $6.30 \leq$ Mass $< 6.60$ | 0.46 | 0.76 |
| $6.60 \leq$ Mass $< 6.90$ | 0.31 | 0.39 |
| $6.90 \leq$ Mass $< 7.50$ | 2.58 | 3.51 |
| $7.50 \leq$ Mass $< 8.80$ | 12.96 | 13.39 |

# Unfolding Systematics: $0.50 \leq x_F < 0.55$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 8.94 | 8.32 |
| $4.20 \leq$ Mass $< 4.50$ | 4.82 | 4.90 |
| $4.50 \leq$ Mass $< 4.80$ | 9.91 | 9.26 |
| $4.80 \leq$ Mass $< 5.10$ | 1.26 | 1.71 |
| $5.10 \leq$ Mass $< 5.40$ | 5.27 | 4.48 |
| $5.40 \leq$ Mass $< 5.70$ | 1.51 | 1.98 |
| $5.70 \leq$ Mass $< 6.00$ | 3.50 | 2.93 |
| $6.00 \leq$ Mass $< 6.30$ | 1.49 | 1.20 |
| $6.30 \leq$ Mass $< 6.60$ | 3.04 | 2.32 |
| $6.60 \leq$ Mass $< 6.90$ | 1.55 | 0.38 |
| $6.90 \leq$ Mass $< 7.50$ | 0.45 | 0.35 |
| $7.50 \leq$ Mass $< 8.80$ | 0.95 | 0.24 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.02 | 7.34 |
| $4.20 \leq$ Mass $< 4.50$ | 3.65 | 2.51 |
| $4.50 \leq$ Mass $< 4.80$ | 4.66 | 5.33 |
| $4.80 \leq$ Mass $< 5.10$ | 2.38 | 2.72 |
| $5.10 \leq$ Mass $< 5.40$ | 0.56 | 0.16 |
| $5.40 \leq$ Mass $< 5.70$ | 0.66 | 0.59 |
| $5.70 \leq$ Mass $< 6.00$ | 1.33 | 1.13 |
| $6.00 \leq$ Mass $< 6.30$ | 0.16 | 0.24 |
| $6.30 \leq$ Mass $< 6.60$ | 0.52 | 0.78 |
| $6.60 \leq$ Mass $< 6.90$ | 3.15 | 3.24 |
| $6.90 \leq$ Mass $< 7.50$ | 1.25 | 1.38 |
| $7.50 \leq$ Mass $< 8.80$ | 9.95 | 10.87 |

# Unfolding Systematics: $0.55 \leq x_F < 0.60$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 6.77 | 7.69 |
| $4.20 \leq$ Mass $< 4.50$ | 3.47 | 3.87 |
| $4.50 \leq$ Mass $< 4.80$ | 6.09 | 6.20 |
| $4.80 \leq$ Mass $< 5.10$ | 0.76 | 0.51 |
| $5.10 \leq$ Mass $< 5.40$ | 1.81 | 1.44 |
| $5.40 \leq$ Mass $< 5.70$ | 0.33 | 0.93 |
| $5.70 \leq$ Mass $< 6.00$ | 2.47 | 2.15 |
| $6.00 \leq$ Mass $< 6.30$ | 0.62 | 0.31 |
| $6.30 \leq$ Mass $< 6.60$ | 2.90 | 3.36 |
| $6.60 \leq$ Mass $< 6.90$ | 4.17 | 4.39 |
| $6.90 \leq$ Mass $< 7.50$ | 1.45 | 1.06 |
| $7.50 \leq$ Mass $< 8.80$ | 10.58 | 10.67 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.28 | 7.54 |
| $4.20 \leq$ Mass $< 4.50$ | 4.30 | 4.49 |
| $4.50 \leq$ Mass $< 4.80$ | 6.29 | 5.86 |
| $4.80 \leq$ Mass $< 5.10$ | 0.19 | 0.92 |
| $5.10 \leq$ Mass $< 5.40$ | 1.92 | 1.35 |
| $5.40 \leq$ Mass $< 5.70$ | 0.54 | 0.19 |
| $5.70 \leq$ Mass $< 6.00$ | 2.56 | 2.12 |
| $6.00 \leq$ Mass $< 6.30$ | 0.43 | 0.54 |
| $6.30 \leq$ Mass $< 6.60$ | 3.24 | 2.92 |
| $6.60 \leq$ Mass $< 6.90$ | 4.07 | 3.89 |
| $6.90 \leq$ Mass $< 7.50$ | 1.24 | 0.86 |
| $7.50 \leq$ Mass $< 8.80$ | 4.79 | 4.42 |
| $8.80 \leq$ Mass $< 10.00$ | 5.31 | 5.60 |

# Unfolding Systematics: $0.60 \leq x_F < 0.65$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 9.53 | 9.74 |
| $4.20 \leq$ Mass $< 4.50$ | 4.87 | 6.04 |
| $4.50 \leq$ Mass $< 4.80$ | 10.02 | 9.71 |
| $4.80 \leq$ Mass $< 5.10$ | 3.03 | 3.44 |
| $5.10 \leq$ Mass $< 5.40$ | 0.74 | 0.62 |
| $5.40 \leq$ Mass $< 5.70$ | 2.57 | 2.72 |
| $5.70 \leq$ Mass $< 6.00$ | 2.10 | 1.77 |
| $6.00 \leq$ Mass $< 6.30$ | 1.23 | 1.14 |
| $6.30 \leq$ Mass $< 6.60$ | 1.14 | 1.30 |
| $6.60 \leq$ Mass $< 6.90$ | 0.22 | 0.56 |
| $6.90 \leq$ Mass $< 7.50$ | 0.73 | 0.50 |
| $7.50 \leq$ Mass $< 8.80$ | 4.95 | 4.90 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 6.69 | 5.91 |
| $4.20 \leq$ Mass $< 4.50$ | 3.03 | 2.47 |
| $4.50 \leq$ Mass $< 4.80$ | 5.19 | 4.92 |
| $4.80 \leq$ Mass $< 5.10$ | 2.24 | 1.77 |
| $5.10 \leq$ Mass $< 5.40$ | 0.70 | 0.29 |
| $5.40 \leq$ Mass $< 5.70$ | 0.21 | 0.72 |
| $5.70 \leq$ Mass $< 6.00$ | 1.37 | 0.76 |
| $6.00 \leq$ Mass $< 6.30$ | 3.29 | 3.37 |
| $6.30 \leq$ Mass $< 6.60$ | 2.43 | 2.68 |
| $6.60 \leq$ Mass $< 6.90$ | 1.46 | 1.44 |
| $6.90 \leq$ Mass $< 7.50$ | 3.59 | 4.00 |
| $7.50 \leq$ Mass $< 8.80$ | 0.34 | 0.32 |

# Unfolding Systematics: $0.65 \leq x_F < 0.70$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.47 | 7.86 |
| $4.20 \leq$ Mass $< 4.50$ | 3.67 | 2.75 |
| $4.50 \leq$ Mass $< 4.80$ | 6.01 | 7.16 |
| $4.80 \leq$ Mass $< 5.10$ | 3.91 | 4.78 |
| $5.10 \leq$ Mass $< 5.40$ | 0.72 | 0.68 |
| $5.40 \leq$ Mass $< 5.70$ | 3.68 | 4.19 |
| $5.70 \leq$ Mass $< 6.00$ | 1.45 | 1.71 |
| $6.00 \leq$ Mass $< 6.30$ | 1.38 | 0.95 |
| $6.30 \leq$ Mass $< 6.60$ | 2.35 | 2.84 |
| $6.60 \leq$ Mass $< 6.90$ | 0.50 | 0.04 |
| $6.90 \leq$ Mass $< 7.50$ | 5.72 | 5.92 |
| $7.50 \leq$ Mass $< 8.80$ | 12.75 | 11.11 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.64 | 8.49 |
| $4.20 \leq$ Mass $< 4.50$ | 4.75 | 4.34 |
| $4.50 \leq$ Mass $< 4.80$ | 6.46 | 8.04 |
| $4.80 \leq$ Mass $< 5.10$ | 2.35 | 2.48 |
| $5.10 \leq$ Mass $< 5.40$ | 0.32 | 1.43 |
| $5.40 \leq$ Mass $< 5.70$ | 3.51 | 3.66 |
| $5.70 \leq$ Mass $< 6.00$ | 2.75 | 2.67 |
| $6.00 \leq$ Mass $< 6.30$ | 1.71 | 2.28 |
| $6.30 \leq$ Mass $< 6.60$ | 2.11 | 3.08 |
| $6.60 \leq$ Mass $< 6.90$ | 2.80 | 3.68 |
| $6.90 \leq$ Mass $< 7.50$ | 3.87 | 4.01 |
| $7.50 \leq$ Mass $< 8.80$ | 5.15 | 5.88 |

# Unfolding Systematics: $0.70 \leq x_F < 0.75$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 5.55 | 6.20 |
| $4.20 \leq$ Mass $< 4.50$ | 3.37 | 4.21 |
| $4.50 \leq$ Mass $< 4.80$ | 3.66 | 4.57 |
| $4.80 \leq$ Mass $< 5.10$ | 2.02 | 1.93 |
| $5.10 \leq$ Mass $< 5.40$ | 2.67 | 1.85 |
| $5.40 \leq$ Mass $< 5.70$ | 0.86 | 0.18 |
| $5.70 \leq$ Mass $< 6.00$ | 3.29 | 3.15 |
| $6.00 \leq$ Mass $< 6.30$ | 7.62 | 7.61 |
| $6.30 \leq$ Mass $< 6.60$ | 2.16 | 1.16 |
| $6.60 \leq$ Mass $< 6.90$ | 15.29 | 15.68 |
| $6.90 \leq$ Mass $< 7.50$ | 16.78 | 17.96 |
| $7.50 \leq$ Mass $< 8.80$ | 10.03 | 11.43 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.27 | 6.99 |
| $4.20 \leq$ Mass $< 4.50$ | 2.55 | 2.08 |
| $4.50 \leq$ Mass $< 4.80$ | 6.61 | 6.26 |
| $4.80 \leq$ Mass $< 5.10$ | 4.63 | 4.93 |
| $5.10 \leq$ Mass $< 5.40$ | 0.27 | 0.64 |
| $5.40 \leq$ Mass $< 5.70$ | 4.44 | 4.37 |
| $5.70 \leq$ Mass $< 6.00$ | 3.01 | 4.23 |
| $6.00 \leq$ Mass $< 6.30$ | 2.60 | 2.56 |
| $6.30 \leq$ Mass $< 6.60$ | 5.70 | 7.16 |
| $6.60 \leq$ Mass $< 6.90$ | 3.69 | 4.86 |
| $6.90 \leq$ Mass $< 7.50$ | 2.27 | 2.17 |
| $7.50 \leq$ Mass $< 8.80$ | 1.63 | 2.19 |

# Unfolding Systematics: $0.75 \leq x_F < 0.80$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 8.78 | 9.16 |
| $4.20 \leq$ Mass $< 4.50$ | 6.09 | 5.54 |
| $4.50 \leq$ Mass $< 4.80$ | 12.66 | 13.38 |
| $4.80 \leq$ Mass $< 5.10$ | 3.71 | 4.40 |
| $5.10 \leq$ Mass $< 5.40$ | 2.97 | 2.87 |
| $5.40 \leq$ Mass $< 5.70$ | 2.71 | 3.19 |
| $5.70 \leq$ Mass $< 6.00$ | 5.32 | 5.11 |
| $6.00 \leq$ Mass $< 6.30$ | 0.79 | 1.29 |
| $6.30 \leq$ Mass $< 6.60$ | 7.34 | 7.41 |
| $6.60 \leq$ Mass $< 6.90$ | 5.86 | 6.14 |
| $6.90 \leq$ Mass $< 7.50$ | 3.79 | 2.37 |
| $7.50 \leq$ Mass $< 8.80$ | 0.04 | 1.53 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 8.43 | 9.00 |
| $4.20 \leq$ Mass $< 4.50$ | 6.03 | 4.81 |
| $4.50 \leq$ Mass $< 4.80$ | 10.66 | 11.52 |
| $4.80 \leq$ Mass $< 5.10$ | 0.93 | 1.49 |
| $5.10 \leq$ Mass $< 5.40$ | 5.41 | 5.10 |
| $5.40 \leq$ Mass $< 5.70$ | 1.71 | 2.09 |
| $5.70 \leq$ Mass $< 6.00$ | 2.27 | 2.09 |
| $6.00 \leq$ Mass $< 6.30$ | 0.66 | 0.76 |
| $6.30 \leq$ Mass $< 6.60$ | 0.64 | 1.07 |
| $6.60 \leq$ Mass $< 6.90$ | 5.78 | 6.79 |
| $6.90 \leq$ Mass $< 7.50$ | 7.78 | 8.94 |
| $7.50 \leq$ Mass $< 8.80$ | 5.45 | 6.44 |

# Unfolding Systematics: $0.80 \leq x_F < 0.85$

## LH2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.88 | 8.37 |
| $4.20 \leq$ Mass $< 4.50$ | 8.83 | 7.87 |
| $4.50 \leq$ Mass $< 4.80$ | 13.07 | 13.90 |
| $4.80 \leq$ Mass $< 5.10$ | 4.26 | 4.72 |
| $5.10 \leq$ Mass $< 5.40$ | 6.51 | 5.72 |
| $5.40 \leq$ Mass $< 5.70$ | 7.01 | 7.54 |
| $5.70 \leq$ Mass $< 6.00$ | 2.76 | 2.08 |
| $6.00 \leq$ Mass $< 6.30$ | 10.21 | 9.48 |
| $6.30 \leq$ Mass $< 6.60$ | 3.20 | 2.26 |
| $6.60 \leq$ Mass $< 6.90$ | 8.58 | 8.85 |
| $6.90 \leq$ Mass $< 7.50$ | 5.07 | 3.81 |
| $7.50 \leq$ Mass $< 8.80$ | 10.41 | 13.76 |

## LD2 Target

| Mass Bin | Syst. Error (Clean) [%] | Syst. Error (Messy) [%] |
|---|---|---|
| $3.90 \leq$ Mass $< 4.20$ | 7.54 | 7.32 |
| $4.20 \leq$ Mass $< 4.50$ | 7.01 | 6.35 |
| $4.50 \leq$ Mass $< 4.80$ | 8.91 | 8.68 |
| $4.80 \leq$ Mass $< 5.10$ | 2.37 | 1.97 |
| $5.10 \leq$ Mass $< 5.40$ | 3.91 | 3.41 |
| $5.40 \leq$ Mass $< 5.70$ | 5.86 | 4.57 |
| $5.70 \leq$ Mass $< 6.00$ | 3.76 | 1.25 |
| $6.00 \leq$ Mass $< 6.30$ | 9.16 | 9.74 |
| $6.30 \leq$ Mass $< 6.60$ | 2.00 | 2.66 |
| $6.60 \leq$ Mass $< 6.90$ | 12.11 | 11.44 |
| $6.90 \leq$ Mass $< 7.50$ | 6.96 | 6.34 |
| $8.80 \leq$ Mass $< 10.00$ | 2.60 | 2.59 |

# Conclusions and Next Steps

- Successfully implemented multi-dimensional unfolding for the double differential cross-section.
- Recalculation of acceptance corrections for Runs 2 and 3 independently is currently underway (Kenichi and Harsha).
- Finalize the publication draft and initiate the collaboration review process.